

Aston University

Visa Compliance for the Future: A Machine Learning Perspective

A dissertation submitted in fulfilment of the requirements for the degree of Master of Science in Data Science.

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Declaration

I declare that all sentences, passages quoted in this report, as well as any data or methodologies referenced, have been explicitly acknowledged with clear citations specifying the source, author, date, and pages. Any part of my own written words based on the work of others is accompanied by proper citations to ensure appropriate credit is given. I understand that failure to properly acknowledge sources constitutes plagiarism and may result in penalties, including failure in this module.

Abstract

In response to the UK Government's 2025 white paper on immigration reform, which tightened visa sponsorship compliance requirements for higher education institutions, this study explores the application of machine learning to support risk assessment in the student visa sponsorship process. Using anonymised application data from Aston University's CASshield system across four intake cycles (2022–2025), comprising 3,957 records with 16 features, this research develops and evaluates predictive models capable of classifying applicants based on credibility interview outcomes.

The study implements a comprehensive supervised learning pipeline, including data preprocessing, exploratory analysis, comparative model evaluation, and systematic hyperparameter optimisation. Following initial evaluation of multiple classification algorithms, Random Forest was selected as the optimal base classifier. A two-stage optimisation approach employing RandomizedSearchCV and GridSearchCV identified an optimal configuration achieving a test set balanced accuracy of 0.8456, representing an improvement of 0.0472 over the baseline model. The optimised classifier demonstrates strong discriminative ability (ROC-AUC: 0.8876) and high sensitivity for the minority "Failed" class (recall: 88.89%), which is critical for compliance risk detection.

Feature importance analysis reveals that a mix of contextual variables have the most discriminative impact on the model (Accommodation, Age, Student Deposit ammount, course fee) are the most predictive features, collectively accounting for over 94% of model importance. Cross-validation results indicate stable generalisation performance, with mean balanced accuracy of 0.8614 ($\pm$ 0.0271) across five folds.

The results demonstrate that machine learning can effectively support visa compliance decision-making by providing early risk indicators for student visa sponsorship. The developed classifier represents a robust proof-of-concept that could inform the development of a production prototype for institutional use. However, model performance could be further enhanced through the integration of additional informative variables and more comprehensive data collection, suggesting that future work should focus on expanding the feature set to improve both predictive accuracy and interpretability for compliance officers.

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Chapter 1

Introduction

1.1 Research Context and Motivation

Immigration has long been a focal point of global political discourse, shaping national policies and influencing public sentiment across generations. In the United Kingdom, migration policy remains a core component of governmental strategy and a persistent topic of public debate.

In May 2025, the UK Government released a white paper titled *Restoring Control Over the Immigration System* (Home Office, 2025). This document formed part of a comprehensive review of all immigration routes into the UK, with a stated aim of reducing net migration and strengthening institutional compliance mechanisms. Among the most affected areas were the higher education and university visa sponsorship systems, which were identified as requiring enhanced oversight and accountability. This study responds directly to these developments by exploring data-driven approaches that could assist universities in maintaining compliance under the new regulatory framework.

Under the current immigration framework, academic institutions may apply for a visa sponsorship licence permitting them to sponsor international students for study in the UK. While this route is critical for the country's global academic engagement, it has often been exploited by individuals whose primary intention may not be genuine study; in comparison to less legal means like small boat crossings or claiming asylum, the student visa route allows for legal working and rental rights in the UK, with a relatively equal economic cost to the applicant. The student visa route constitutes approximately half of all non-visit visas, facilitating the migration of over 400,000 annually (Office for National Statistics, 2024).

Academic institutions vary greatly in their approach to recruiting students, and monitoring their applicants for their genuine intention to study (Onk, 2017; Huang, 2016). As the wider higher education sector in the UK faces financial pressure due to declining student numbers, and stagnant home tuition fees, many institutions choose to recruit from abroad where students can pay an uncapped tuition fee and often pay 3-4 times more than an equivalent domestic student. Because of these sector wide financial pressures and the increased financial incentive to recruit from abroad, there is fierce competition in the market for international students, and some institutions have already been observed to lower their entry requirements for international students only. Opportunistic economic

migrants often try to exploit this financial desperation, by submitting applications to study with false documents, and a small deposit (Often between £1000-£2000) in order to gain student visa sponsorship (Highet, 2025). After arriving in the UK, applicants on a student visa have the right to work 20 hours per week inside of term time and rent privately, something which asylum seekers do not get. This then inevitably leads to many non-genuine students flouting their academic responsibilities to pursue relatively well paid work in the UK for as long as they can.

A visa sponsorship license comes with many responsibilities and obligations to prevent abuse of the system, chiefly, all sponsors must successfully complete an annual Basic Compliance Assessment (BCA) which measures key metrics of concern to the government. Before the white paper was published, institutions were given a 10% threshold for visa refusals of sponsored applicants, this was to ensure that institutions took responsibility for their sponsee's visa applications and took measures to ensure only credible students were sponsored. In response to the white papers findings, the government has changed this threshold from 10% to 5%, effectively halving the room for error in sponsorship risk. Other metrics have also been changed to be "tighter" such as the course completion rate for sponsored students and their enrolment rate.

These regulatory changes force universities to reevaluate their approaches to visa sponsorship and international recruitment or face the possibility of losing the visa sponsorship privilege altogether.

1.2 Aims and Objectives

The primary aim of this project is to develop and evaluate machine learning models capable of identifying compliance risk within the student visa sponsorship process. Using real-world data from previous international student intakes at Aston University, the study will test a range of algorithmic approaches to classification and risk prediction.

Specifically, the project will:

1. Build a predictive model classifying applicants based on visa outcome or credibility interview results.
2. Compare the performance of different ML algorithms to determine the most effective approach for risk classification.
3. Assess model interpretability and real-world applicability in a university compliance setting.

Ultimately, the study seeks to determine whether ML can support institutional decision-making by offering an early-warning mechanism for sponsorship risk.

1.3 Research Questions

This research seeks to address the following questions:

- Is the data collected by Aston University sufficient to develop an accurate and reliable risk classification model?
- Can a prototype classifier be developed that generalises effectively to unseen applicant data?
- Can this model be implemented into Aston Universities Visa Compliance strategy and what benefit could this bring?

Chapter 2

Literature Review

2.1 Article Selection

This literature review focuses on papers published between 2010 to 2025, focusing on the application Machine Learning(ML) in risk and compliance environments. All sources were obtained from Google Scholar, and provide an overview of the relevant technology needed for this project.

2.2 Foundations and Theoretical Background

Machine learning is a broad field of artificial intelligence concerned with the development of algorithms that enable computers to learn patterns from data and improve performance on a task without being explicitly programmed. As defined in the literature, ML builds upon statistical learning theory, optimisation methods, and computational pattern recognition. Early foundational work, such as that by Vapnik (1999) on statistical learning, established core principles including empirical risk minimisation and the importance of balancing model complexity against predictive accuracy. These ideas underpin modern ML practice, where the objective is to identify a function that maps inputs to outputs in a way that generalises effectively to unseen data.

Gupta (2022) outlines how the field is commonly divided into supervised, unsupervised, and reinforcement learning. Supervised learning involves training a model on labelled examples to predict discrete categories or continuous values. Algorithms such as linear regression, logistic regression, decision trees, random forests, and neural networks have become widely adopted due to their strong empirical performance and theoretical grounding. Unsupervised learning, by contrast, seeks to discover hidden structure in unlabelled data through techniques such as clustering, dimensionality reduction, and density estimation. Reinforcement learning focuses on agents that learn optimal behaviour through interaction with an environment, guided by reward signals.

A unifying characteristic across these paradigms is the reliance on optimisation. Most ML models aim to minimise a loss function that quantifies the discrepancy between predicted and actual outcomes. Gradient-based optimisation techniques including stochastic gradient descent and its variants have enabled the training of complex, high-dimensional

models, particularly deep neural networks. Advances in computational power, data availability, and algorithmic innovation have further accelerated ML's growth, allowing models to capture nonlinear relationships, interactions, and latent structures that traditional statistical methods may miss.

Papers like Barbiero (2020) also emphasise the importance of evaluation and generalisation. Metrics such as accuracy, precision, recall, F-score, and area under the ROC curve are standard tools for assessing model performance, while cross-validation, regularisation, and model selection frameworks help guard against overfitting. Together, these concepts form the theoretical foundation upon which contemporary machine learning research and applications are built.

2.3 Foundational Papers on ML in Fraud Detection

Fraud detection represents one of the earliest and most widely used applications of Machine Learning in real-world risk management. Due to large datasets and growing demand, this field has long served as a proving ground for different algorithmic approaches that can be transferred to other compliance-based contexts. The papers discussed below cover broad conceptual and technical foundations of automated fraud detection and define common challenges such as class imbalance, evolving behaviour patterns, false positives and missed detections.

Phua et al. (2010) was a cornerstone publication which grounds the basis for ML is fraud detection application. This paper serves to highlight a lot of the specific ML challenges within a fraud detection context: heavily unbalanced datasets, high misclassification impact, concept drift, cost sensitive learning, etc. This is an important paper for this study, as it proves that many different ML approaches may be suitable for risk or compliance detection within the specific visa context of this study.

Ngai et al. (2011) further refines techniques for fraud detection and explores data-mining techniques in a financial fraud context. They also provide a systematic framework for comparing the viability or efficacy of different ML models which focuses on accuracy, scalability and interpretability, this is something which will be used to compare ML models within this study. This study also seems to suggest that no single model is universally optimal, and that hybrid and ensemble models are typically the strongest.

A further (more recent) literature review from Hernandez Aros et al. (2024) fast tracks us through over a decade of advancement within the field of fraud detection with machine learning. This review focuses on studies published from 2012 to 2023, and identifies a clear shift from feature engineering driven pipelines to end to end architectures such as recurrent or graph neural networks, which can model complex temporal or relational dependencies in financial transactions. What is also of supreme relevance within this study is it's noting of the growing need for explainability within ML solutions, as many modern institutions must be able to justify algorithmic decisions to relevant regulatory and compliance authorities.

A further relevant study from Gama et al. (2014) focuses on concept drift adaptation. This paper addresses the challenge of declining performance from static ML models that operate within a dynamic and adversarial environment. Essentially this paper proposes

an array of approaches which improve a model's adaptability, this may be something we are able to test and implement in our study, as we have many years of data to work with. This paper also touches on the ongoing nature of risk modelling and how detection algorithms must be periodically retrained to reflect changing policy, behaviour, geopolitical circumstance etc.

Collectively, these foundational works established the methodological backbone of ML based fraud detection. They highlight recurring themes that directly inform modern compliance automation: the need for adaptive, interpretable models; the handling of severe class imbalance; and the continuous evaluation of model performance under changing conditions. As this dissertation seeks to apply similar principles to visa-sponsorship compliance, the lessons from financial fraud detection, balancing predictive accuracy with accountability provide both a conceptual and practical blueprint for responsible innovation in public-sector risk assessment.

2.4 ML for Visa Application Assessment

In recent years, governments and researchers have begun exploring how data-driven technologies, including machine learning (ML), can support immigration and visa decision-making. The underlying logic mirrors that used in financial risk analysis predictive algorithms can identify patterns of non-compliance or deception within large datasets and improve consistency in administrative decision-making. However, this emerging field remains contested, balancing potential efficiency gains against ethical and legal constraints.

A foundational empirical contribution comes from Chen et al. (2017), who examined whether asylum adjudication outcomes in the United States could be predicted using machine learning. Their study framed judicial asylum decisions as a classification problem, incorporating features such as applicant nationality, legal representation, and historical behaviour. Logistic regression, random forest, and support vector machine (SVM) models achieved predictive accuracies exceeding 80%, suggesting that asylum outcomes follow statistically learnable patterns rather than being purely discretionary. The work of Chen et al. (2017) demonstrated both the promise and the sensitivity of predictive modelling in migration contexts: while ML can identify hidden determinants of case outcomes, it also risks entrenching systemic biases embedded in historical data.

Complementing these empirical studies, policy-oriented analyses such as the paper by Graham (2021) on "Automating Decision-Making in Migration Policy" have mapped the growing integration of automated systems across migration management. The paper documents how states including the UK, Canada, and New Zealand have adopted algorithmic tools to support visa triaging, document verification, and identity fraud detection. These systems often function as decision-support rather than decision-making tools, providing risk scores or confidence levels to assist human officers. Graham highlights how such automation can enhance consistency and resource efficiency but warns that opacity in algorithmic design risks undermining public trust and due process. The report calls for transparent governance frameworks, human oversight, and periodic auditing—principles directly relevant to any ML system designed for visa sponsorship compliance.

2.5 Research Gap and Study Contribution

While the existing literature provides valuable insights into machine learning applications for fraud detection and government-level visa decision-making, a significant gap remains in the application of ML to institutional-level visa sponsorship compliance within higher education contexts. The studies reviewed above focus primarily on two domains: financial fraud detection in commercial settings, and government-level immigration decisions (such as asylum adjudications or national visa processing systems). However, the specific context of university visa sponsorship compliance where institutions act as intermediaries responsible for assessing applicant credibility before government visa decisions has received limited attention in the machine learning literature.

This gap is particularly relevant given the evolving regulatory landscape. The UK Government's 2025 white paper on immigration reform has tightened compliance requirements for higher education institutions, reducing the visa refusal threshold from 10% to 5% and increasing accountability for sponsored applicants. This regulatory shift creates an urgent need for institutions to develop more effective risk assessment mechanisms, yet there is little published research on how machine learning can support this specific institutional responsibility. Unlike government-level systems that process final visa decisions, university compliance teams must assess credibility earlier in the process, using different data sources (application materials, interviews, financial documentation) and operating under different constraints (maintaining recruitment targets while ensuring compliance).

This study addresses this gap by developing and evaluating machine learning models specifically for institutional visa sponsorship compliance, using real-world data from a UK university's credibility interview process. The research contributes to the literature by: demonstrating the applicability of ML techniques to the unique context of university-level compliance assessment, evaluating model performance on institutional data that differs from government-level datasets, and providing insights into how interpretable ML models can support compliance officers in their decision-making processes. By bridging the gap between general fraud detection methodologies and the specific needs of higher education visa compliance, this study offers both practical value for institutions facing tightened regulatory requirements and theoretical contributions to the growing field of ML applications in public-sector risk assessment.

Chapter 3

Data

This chapter outlines the dataset used in the study, including its origin, structure, anonymisation procedures, ethical considerations, and key characteristics relevant to subsequent modelling stages. A systematic overview of the dataset is provided, followed by a description of the features retained for analysis and the limitations inherent in the data. This chapter concludes with a summary of initial preprocessing steps conducted prior to model development.

3.1 Overview

The dataset used in this research was obtained from Aston University's Visa Compliance Team and is derived from a generalised report used internally to monitor the progress of international applicants. The report is generated through the University's CASshield system, which is used to evaluate applicants for issuance of a Confirmation of Acceptance for Studies (CAS).

The dataset comprises application records from the September 2022, 2023, 2024, and 2025 intake cycles. Across these years, CASshield captures approximately 35 variables, although the exact features available vary slightly by application cycle. Data extraction and initial handling were undertaken within the Visa Compliance Team to ensure secure and compliant data transfer for research purposes.

All data used in this project was anonymised prior to analysis, and the study received explicit approval from both the Visa Compliance Team and the Academic Registrar. Only information directly relevant to the modelling task was retained, with all personal identifiers removed or transformed to comply with data-protection requirements.

3.2 Anonymization

Because the dataset contained real information about international applicants, extensive anonymisation was required to ensure that individuals could not be identified and that all processing complied with the Data Use and Access Act (2025).

Prior to its release for research analysis, the Visa Compliance Team applied a series of irreversible transformations within the secure operational environment of University systems. Direct identifiers including names, date of birth, email addresses, phone numbers, and other personal data fields were removed entirely. Certain identifying attributes that held analytical relevance, such as first and last names, were replaced with pseudonymised proxy tokens that preserved structural relationships in the data while preventing any connection to real individuals.

Several features that could indirectly reveal confidential outcomes or provide downstream hints about an applicant's immigration status, such as visa status indicators or sharecodes, were excluded from the extracted dataset to avoid leakage into the modelling process. Administrative variables with no relevance to the predictive task were also removed to reduce noise.

Interview outcome was selected as the target variable for the purposes of this project, any students that were not required to complete a Pre-CAS Interview were removed from this dataset.

Following these anonymisation procedures and the subsequent selection of analytically relevant attributes, a refined set of variables was retained for use in the modelling component of the study. A full description of these features is provided in Section 3.3.

3.3 Dataset Summary

The final anonymised dataset comprises application records drawn from the September 2022, 2023, 2024, and 2025 intake cycles. These records were combined into a single consolidated dataset for analysis. The total number of records included in the study is 3957, representing the complete set of applicants processed through CASshield during the selected intake periods for whom relevant variables were available. After anonymisation and feature pruning, the dataset contains sixteen features, including the target variable representing the outcome of the credibility interview.

Feature	Type	Description
Accommodation	Categorical	Indicates whether the applicant has arranged accommodation.
Agent Company Name	Categorical	Education agent or company through which the application was submitted.
Course Fee After Scholarships	Numerical	Final tuition fee after financial discounts or scholarships are applied.
Course	Categorical	Academic programme to which the applicant applied.
Course Scholarships Applied	Numerical	Indicates whether the applicant applied for a scholarship.
Country of Application	Categorical	Country from which the applicant submitted their application.
CourseFee	Numerical	Total tuition fee before any reductions.
Dependant Children	Binary	Number of dependent children declared by the applicant.
Dependant Partner	Binary	Indicates whether the applicant declared a dependent partner.
Nationality	Categorical	Applicant's stated nationality.
Student Deposit Amount	Numerical	Tuition deposit amount paid at the point of CAS consideration.
Age(years)	Numerical	Age of the applicant at the time of application.
Application Year	Categorical	Intake cycle corresponding to the application record.
First Name Proxy	Categorical	Pseudonymised identifier derived from the applicant's first name.
Last Name Proxy	Categorical	Pseudonymised identifier derived from the applicant's surname.
Interview Outcome	Binary	Target variable representing the result of the credibility interview.

Table 3.1: Features included in the anonymised dataset.

This dataset was mostly complete, with very few missing values; only 4.19% of the dataset contained missing values, and the features missing information were exclusively "Accommodation" and "Agent Company Name" which is a result of students not having used these services.

3.4 Data Limitations and Considerations

Several limitations associated with the dataset must be acknowledged. Variations in the CASshield reporting structure across the 2022–2025 period mean that certain features were not consistently available for every intake cycle, potentially introducing structural variability. The dataset also reflects the operational procedures and applicant population of a single UK higher education institution; therefore, findings derived from this analysis may not directly generalise to other universities or national contexts.

The substitution of names with pseudonymised proxies, while essential for privacy protection, removes cultural or linguistic cues that might otherwise contribute subtle predictive signals. In addition, the dataset is heavily unbalanced, with 400 failure outcomes and 3557 pass outcomes, strategies to account for this will be further detailed in the Methodology section.

3.5 Ethical Considerations

The use of applicant data in this study required careful attention to privacy, data-protection responsibilities, and proportionality of processing. All data was obtained with explicit institutional approval, and the extraction, anonymisation, and transfer procedures were carried out by the Visa Compliance Team to ensure compliance with the Data Use and Access Act (2025).

Only variables necessary for the analytical aims of the project were retained, and direct or indirect identifiers were removed or pseudonymised to prevent re-identification. The dataset was used exclusively for academic research purposes and stored within secure University systems throughout the project. No attempt was made to link records to individual applicants, and no decisions affecting real applicants were informed by this research.

These measures ensured that the handling and analysis of the data remained proportionate, privacy-preserving, and aligned with established ethical and legal obligations.

Chapter 4

Methods

4.1 Overview of Methodological Approach

This study followed a supervised machine learning pipeline designed to develop a predictive model capable of classifying international student applicants according to their credibility interview outcome. The methodological approach consisted of four main stages: (4.2) data preprocessing and integration, (4.3) exploratory data analysis, (4.4) model selection, and (4.5) model optimisation and evaluation. The overall objective was to identify a classification model that generalises effectively to unseen applicants while maintaining robustness to the unbalanced nature of the dataset across multiple years worth of intakes. All code was implemented in Python using Jupyter notebook environments, with random seeds set to 42 throughout to ensure reproducibility.

4.2 Data Preprocessing

All preprocessing operations were conducted on the anonymised dataset described in Chapter 3. Because the raw data originated from multiple intake cycles (2022–2025), the first step involved merging these files into a unified dataset. Structural alignment was required to ensure consistent naming conventions, categorical label formats, and data types.

Features deemed irrelevant to credibility assessment, as well as those considered sensitive or potentially capable of revealing personal identifiers, were removed prior to analysis. The retained features were those identified as relevant for modelling based on domain knowledge and the criteria outlined in Chapter 3.

Missing values were handled through imputation strategies appropriate to each feature type. For numerical features, missing values were imputed using the median strategy, which preserves the distribution of numeric variables while handling the small proportion of missing data (4.19%). The median was chosen over the mean because it is more robust to outliers, which are present in financial features such as Student Deposit Amount and Course Fee. Additionally, the median preserves the central tendency of skewed distributions without being unduly influenced by extreme values, making it more appropriate for financial and demographic variables that may exhibit non-normal distributions. For cat-

egorical features, missing values were imputed using a constant strategy with the value "missing", treating missingness as an informative category. This approach was chosen because exploratory data analysis (see Section 4.3) revealed that missingness in categorical features, particularly Accommodation and Agent company name, was statistically associated with the target variable, indicating that the absence of information itself may be predictive of interview outcomes. Treating missingness as a distinct category preserves this potentially informative signal, rather than discarding it through deletion or potentially introducing bias through mode imputation. This approach was implemented using `SimpleImputer` from scikit-learn (Pedregosa et al., 2011) within the preprocessing pipeline.

Categorical variables were encoded using one-hot encoding via `OneHotEncoder`, which creates binary indicator variables for each category level. This encoding method was chosen to avoid imposing ordinal relationships on categorical features that lack inherent ordering, such as Country, Nationality, and Course, where no meaningful ranking exists between categories. The encoder was configured with `handle_unknown='ignore'` to effectively handle unseen categories in the test set, ensuring that the model can process new data without errors when encountering categories not present during training. The `max_categories=50` parameter was set to limit the dimensionality expansion for high-cardinality features, preventing excessive feature space growth that could lead to overfitting or computational inefficiency, while still preserving sufficient granularity for features with moderate cardinality. Numerical variables were standardized using `StandardScaler` to ensure features were on comparable scales, which is particularly important for distance-based algorithms (such as K-Nearest Neighbors) and neural networks, though tree-based algorithms like Random Forest are less sensitive to feature scaling. Standardization was retained in the pipeline to ensure consistent preprocessing across all evaluated models and to support potential future use of algorithms that require scaled features.

The dataset was split into training and test sets using an 80/20 split ratio, with stratification applied to maintain the class distribution across both sets. The 80/20 ratio was chosen as a standard practice in machine learning that provides sufficient training data (3,165 samples) for model development while reserving an adequate test set (792 samples) for reliable performance evaluation. This ratio balances the competing needs of maximising training data for model learning and maintaining a test set large enough to provide statistically meaningful performance estimates. Stratification was applied to ensure that both the training set (80%) and test set (20%) preserve the approximately 8.89:1 class imbalance observed in the full dataset, preventing evaluation bias that could occur if the test set had a different class distribution than the training set. The split was performed using `train_test_split` from scikit-learn (Pedregosa et al., 2011) with `random_state=42` to ensure reproducibility.

No additional feature engineering was applied beyond type harmonisation and the preprocessing steps described above, as initial exploratory analysis indicated that existing features contained sufficient discriminatory signal for classification modelling.

4.3 Exploratory Data Analysis

An exploratory data analysis (EDA) was performed on the combined dataset of student interview outcomes across multiple academic years (2022–2025). The dataset contained 3,957 records with 16 features after combining and preprocessing individual year datasets. The analysis focused on students who completed video interviews, filtering to binary outcomes of "Passed" or "Failed" to support classification.

The target variable showed significant class imbalance, with 3,557 students passing (89.89%) and 400 failing (10.11%), representing an 8.89:1 imbalance.

Statistical significance testing was conducted to identify features associated with interview outcomes. For numeric features, one-way ANOVA F-tests identified four significant features ($p < 0.05$): CourseFee, Student Deposit Amount, Uni Deposit Amount, and age. For categorical features, chi-square tests of independence identified nine significant associations ($p < 0.05$), including, first_name_proxy, last_name_proxy, Course, and Agent company name. The high-cardinality anonymised identity proxies first_name_proxy: 1,234, last_name_proxy: 1,115) showed strong associations, though their predictive value may reflect underlying patterns rather than direct causality.

Temporal analysis across dataset years (2022–2025) revealed variations in class distribution. The failure rate ranged from 5.20% (2024) to 12.41% (2023), suggesting temporal shifts in outcomes. The dataset distribution across years was: 2022 (11.3%), 2023 (38.5%), 2024 (22.8%), and 2025 (27.3%). The target variable distribution by year is visualised in Figure 4.1.

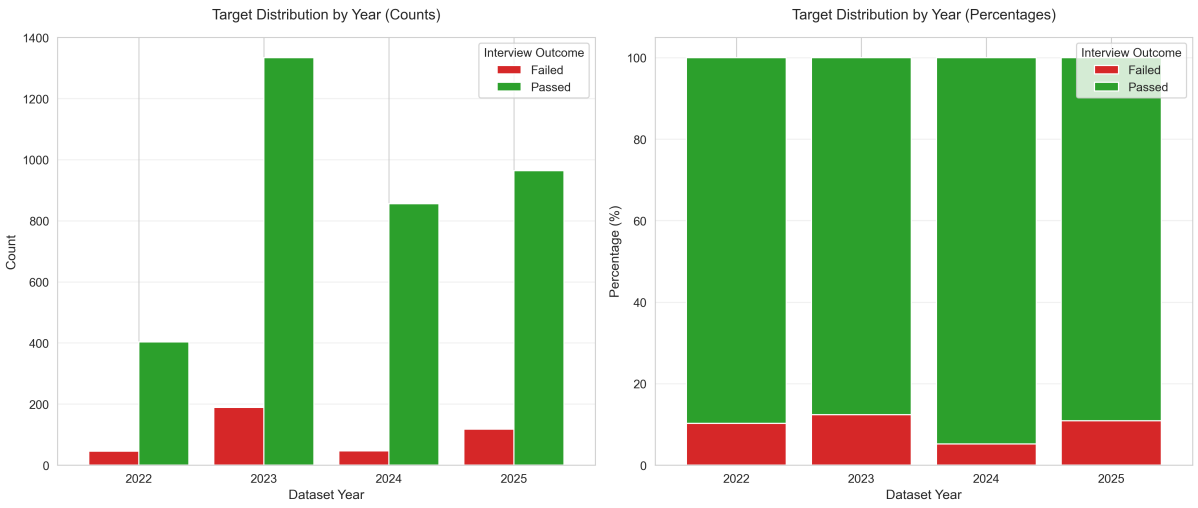


Figure 4.1: Distribution of interview outcomes (Passed/Failed) by dataset year. The stacked bar chart shows both counts and percentages, illustrating temporal variations in failure rates across the four intake cycles (2022–2025).

Correlation analysis of numeric features found no highly correlated pairs ($|r| > 0.7$), indicating low multicollinearity. Moderate correlations included Course Fee After Scholarships with CourseFee ($r = 0.466$) and Uni Deposit Amount with Course Scholarships Applied ($r = 0.538$), which were expected given their conceptual relationships. The correlation matrix for all numeric features is visualised in Figure 4.2.

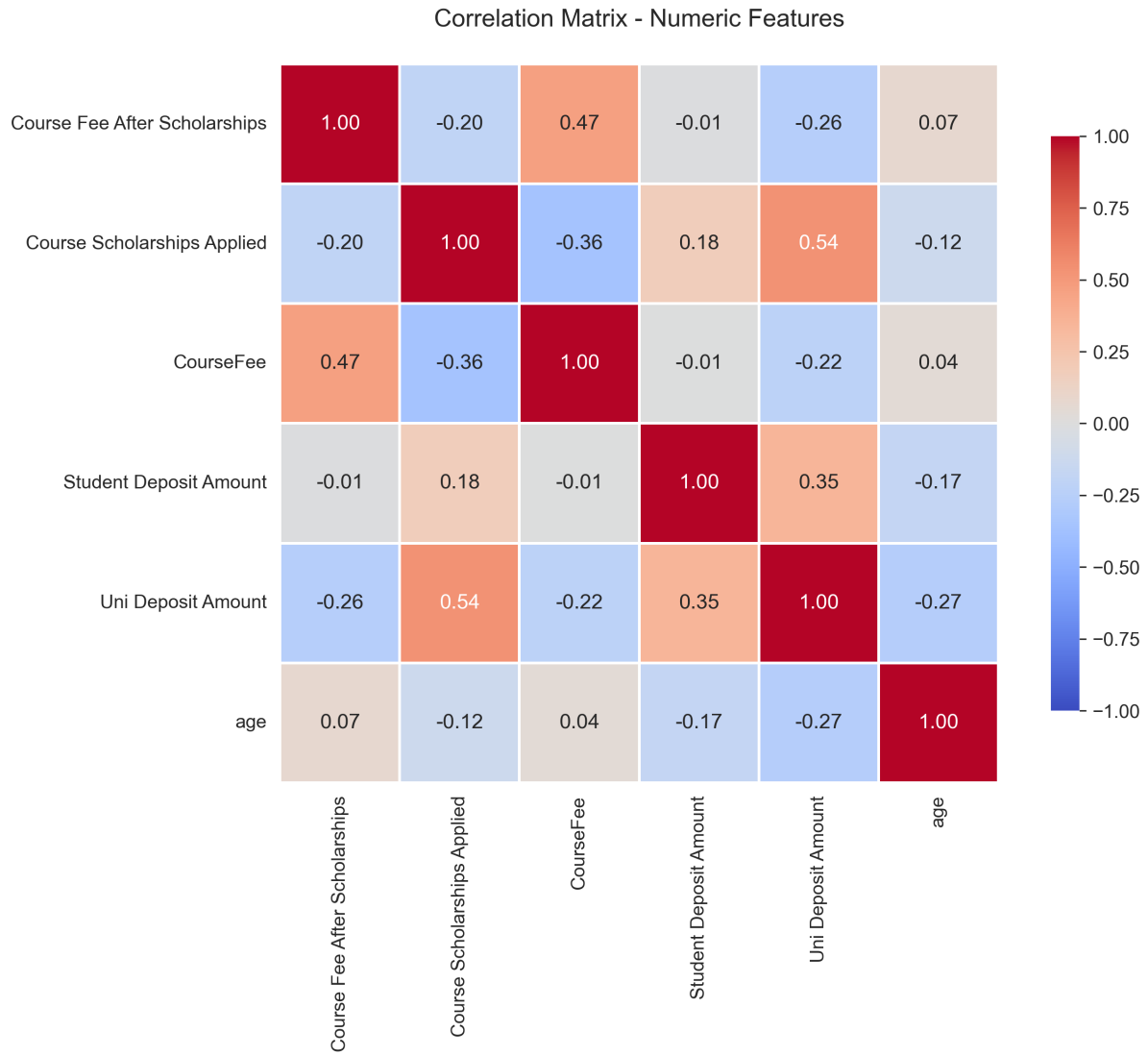


Figure 4.2: Correlation heatmap of numeric features in the combined dataset. Values range from -1 (perfect negative correlation) to +1 (perfect positive correlation), with darker shades indicating stronger correlations.

The analysis identified 14 features with statistically significant associations with interview outcomes, providing a foundation for predictive modeling. The combination of financial, demographic, and temporal features, along with the observed class imbalance and temporal variations, establishes the dataset as suitable for binary classification modeling.

4.4 Model Selection

Model selection was conducted through a systematic comparative evaluation of multiple classification algorithms to identify the most suitable approach for the imbalanced binary classification task. An initial Multi-Layer Perceptron (MLP) neural network was tested with a two-layer architecture (64–32 neurons), L2 regularization ($\alpha = 0.001$), and early stopping to mitigate overfitting. However, the MLP demonstrated poor performance with very low balanced accuracy, indicating it was not an effective classifier for this dataset.

Subsequently, six additional classifiers were evaluated: Decision Tree, Random Forest (Breiman, 2001), Gradient Boosting, Logistic Regression, K-Nearest Neighbors, and AdaBoost. All models were configured with class weighting strategies (specifically `class_weight='balance`) to address the observed 8.89:1 class imbalance. This approach automatically adjusts class weights inversely proportional to class frequencies to prevent the model from being biased toward the majority class (He and Garcia, 2009). Each classifier was evaluated using 5-fold stratified cross-validation on the training set with `random_state=42` for reproducibility (Kohavi, 1995). The 5-fold approach was chosen as a standard cross-validation strategy that provides a good balance between computational efficiency and reliable performance estimation, with each fold containing approximately 20% of the training data, ensuring robust evaluation while maintaining sufficient training data in each fold.

Balanced accuracy was selected as the primary selection metric because it accounts for class imbalance by averaging the recall obtained on each class, making it more appropriate than standard accuracy for imbalanced datasets where the minority class ("Failed") is of primary interest for risk detection. Additional metrics including F1-score (macro and weighted), ROC-AUC, precision, and recall were computed to provide comprehensive performance assessment across different evaluation perspectives. All models were integrated into scikit-learn (Pedregosa et al., 2011) pipelines that combined the preprocessing steps (imputation, encoding, scaling) with the classifier, ensuring consistent preprocessing across all model evaluations and preventing data leakage between training and validation sets.

The comparative evaluation revealed significant performance differences across classifiers. Decision Tree achieved the highest cross-validation balanced accuracy (0.7981), followed by Random Forest (0.6338) and Logistic Regression (0.5946). Gradient Boosting, AdaBoost, and K-Nearest Neighbors demonstrated lower performance, with K-Nearest Neighbors failing to produce valid predictions. The top three performing classifiers (Decision Tree, Random Forest, and Logistic Regression) were further evaluated on the held-out test set. Random Forest achieved the highest test set balanced accuracy (0.7984), outperforming Decision Tree (0.7432) and Logistic Regression (0.7632), and was therefore selected for further optimisation.

4.5 Model Optimisation and Evaluation

Following the selection of Random Forest as the optimal base classifier, systematic hyperparameter optimisation was conducted using a two-stage approach. First, RandomizedSearchCV was employed with 50 iterations to explore a broad parameter space, including the number of estimators (50–300), maximum depth (10–25 or unlimited), minimum samples for split (10–40), minimum samples per leaf (5–20), maximum features (sqrt, log2, or all), and class weighting strategies (balanced or balanced_subsample). This randomised search approach has been shown to be more efficient than exhaustive grid search for hyperparameter optimisation (Bergstra and Bengio, 2012). The 50 iterations were chosen to provide a reasonable exploration of the parameter space while maintaining computational efficiency, as exhaustive grid search across all parameter combinations would be computationally prohibitive. The parameter ranges were selected based on common practices for Random Forest classifiers and the dataset size, with the number of estimators range (50–300) allowing exploration from smaller, faster models to larger,

potentially more accurate ensembles, and depth limits (10–25) preventing excessive tree depth that could lead to overfitting. The best parameters from this initial search achieved a cross-validation balanced accuracy of 0.8197.

The optimal parameters identified through RandomizedSearchCV were then used as a starting point for a refined GridSearchCV, which performed an exhaustive search within a focused parameter space around these values. This second stage tested 576 parameter combinations, identifying a configuration that achieved a cross-validation balanced accuracy of 0.8656, representing an improvement of 0.0459 over the initial randomised search. The final optimised Random Forest model was evaluated on the test set, achieving a balanced accuracy of 0.8456, which represented an improvement of 0.0472 over the baseline Random Forest configuration (balanced accuracy of 0.7984). This two-stage optimisation approach balanced computational efficiency with thorough parameter exploration, ultimately identifying the optimal Random Forest configuration for the classification task.

The final optimised Random Forest model was configured with the following hyperparameters: `n_estimators=300`, `max_depth=15`, `min_samples_split=30`, `min_samples_leaf=20`, `max_features=None` (using all features for each split), and `class_weight='balanced_subsample'` (applying balanced class weights to each bootstrap sample). The model was trained with `random_state=42` and `n_jobs=-1` for parallel processing. This configuration represents the optimal balance between model complexity and generalisation performance, with the increased number of estimators (300 vs. 100 in the baseline) and stricter splitting criteria (min_samples_split=30, min_samples_leaf=20) helping to reduce overfitting while maintaining predictive power on the imbalanced dataset.

Chapter 5

Results

This chapter presents the performance results of the optimised Random Forest classifier on the test set, including comprehensive evaluation metrics, classification performance analysis, and feature importance insights. All results are based on the final model configuration described in Chapter 4, evaluated on the held-out test set comprising 20% of the combined dataset.

5.1 Final Model Performance

The optimised Random Forest classifier was evaluated on the test set (792 samples) following the 80/20 stratified split. Table 5.1 presents the comprehensive performance metrics across multiple evaluation measures.

Metric	**Value**
Balanced Accuracy	0.8456
Accuracy	0.9171
F1-Score (Macro)	0.6006
F1-Score (Weighted)	0.9298
ROC-AUC	0.8876
Precision (Macro)	0.5917
Recall (Macro)	0.8456

Table 5.1: Test set performance metrics for the optimised Random Forest classifier.

The model achieved a balanced accuracy of 0.8456 on the test set, representing an improvement of 0.0472 over the baseline Random Forest configuration (balanced accuracy of 0.7984). The ROC-AUC score of 0.8876 indicates strong discriminative ability, demonstrating that the model can effectively distinguish between "Passed" and "Failed" interview outcomes.

Cross-validation performance on the training set (3,165 samples) showed consistent re-

sults, with a mean balanced accuracy of 0.8197 (± 0.0237) across 5-fold stratified cross-validation, indicating good generalisation and stability of the model.

5.2 Classification Performance Analysis

The confusion matrix for the test set predictions is presented in Figure 5.1. The model demonstrates high sensitivity for the minority class ("Failed"), correctly identifying the majority of actual failures, while maintaining strong performance on the majority class. The confusion matrix shows the distribution of true positives, true negatives, false positives, and false negatives across both classes.

	Failed	**Passed**
Failed	73	7
Passed	135	577

Figure 5.1: Confusion matrix for the optimised Random Forest classifier on the test set. Rows represent actual classes, columns represent predicted classes.

Per-class performance metrics are detailed in Table 5.2. The model shows high precision for the "Passed" class (0.9928), meaning that when the model predicts a student will pass, it is correct 99.28% of the time. For the "Failed" class, precision is lower (0.1905), reflecting the challenge of identifying failures in an imbalanced dataset. However, the high recall for "Failed" (0.8889) indicates that the model successfully identifies the majority of actual failures, which is critical for risk detection in a compliance context.

Class	**Precision**	**Recall**	**F1-Score**
Failed	0.1905	0.8889	0.3137
Passed	0.9928	0.8023	0.8875

Table 5.2: Per-class performance metrics for the optimised Random Forest classifier on the test set.

The classification performance demonstrates a trade-off between precision and recall for the minority class, which is characteristic of imbalanced classification problems. The model prioritises recall for the "Failed" class, ensuring that most actual failures are identified, while accepting a higher rate of false positives where "Passed" students are incorrectly predicted as "Failed".

5.3 Feature Importance and Interpretability

Feature importance analysis was conducted to identify which features contribute most significantly to the model's predictions. The top five most important features, ranked by their importance scores, are presented in Table 5.3.

Rank	Feature	Importance
1	Accommodation (missing)	0.7317
2	Age	0.0745
3	Student Deposit Amount	0.0536
4	Course Fee After Scholarships	0.0349
5	CourseFee	0.0229

Table 5.3: Top five most important features for the optimised Random Forest classifier, ranked by feature importance score.

The most important feature is the accommodation status indicator, specifically whether accommodation information is missing, which accounts for 73.17% of total feature importance. This suggests that applicants who have not arranged accommodation at the point of CAS consideration may represent a higher risk category, potentially indicating less preparation or commitment to the study plan. The distribution of accommodation status by interview outcome is visualised in Figure 5.2.

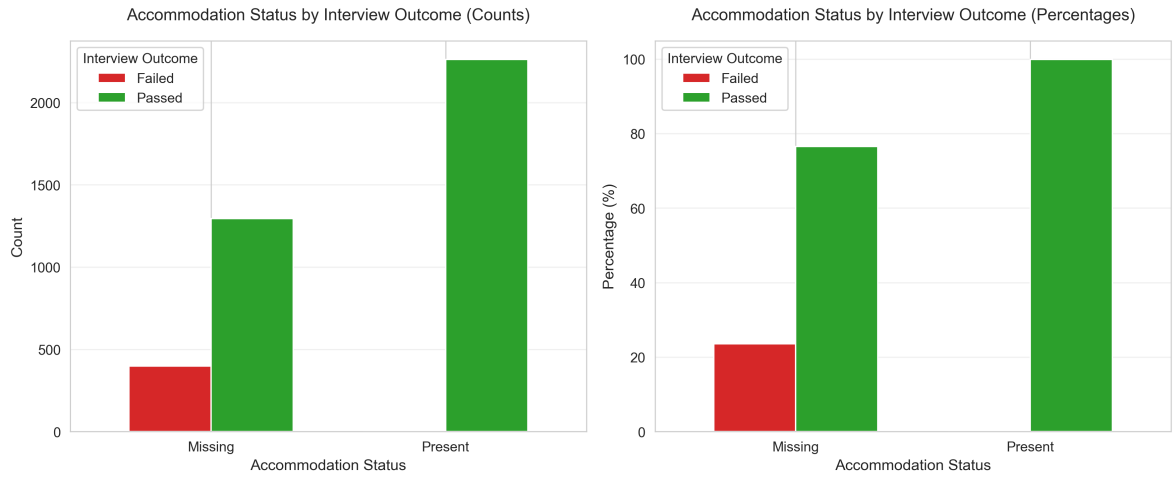


Figure 5.2: Distribution of accommodation status by interview outcome (Passed/Failed). The bar chart illustrates the relationship between accommodation arrangements and interview outcomes, showing the proportion of applicants with and without accommodation information in each outcome class.

Demographic and financial features follow in importance. Age accounts for 7.45% of feature importance, suggesting that applicant age may correlate with interview outcomes, potentially reflecting maturity, preparedness, or other age-related factors relevant to credibility assessment. Financial indicators (Student Deposit Amount: 5.36%, Course Fee After Scholarships: 3.49%, and CourseFee: 2.29%) collectively represent 11.14% of total importance. This aligns with domain knowledge suggesting that financial indicators are relevant to credibility assessment, as applicants with appropriate financial arrangements may demonstrate greater commitment to genuine study. The distribution differences for the top numeric features (Age and Student Deposit Amount) between interview outcome classes are visualised in Figure 5.3, demonstrating how these features differ between Passed and Failed applicants.

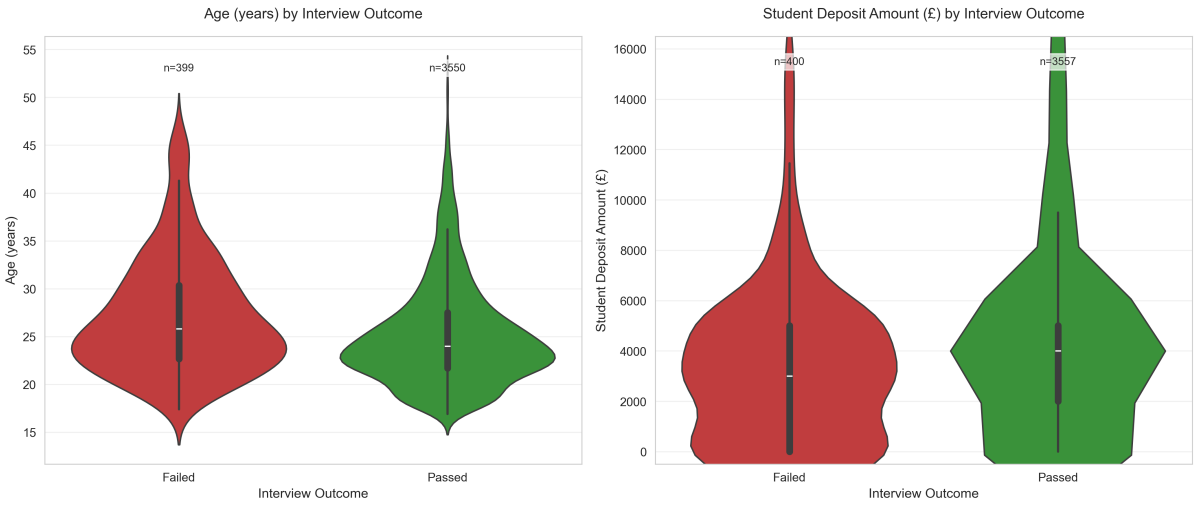


Figure 5.3: Violin plots showing the distribution of Age and Student Deposit Amount by interview outcome (Passed/Failed).

The dominance of the accommodation missing indicator in feature importance provides an interpretable and actionable signal for compliance officers. The combination of accommodation status, age, and financial features among the top predictors offers a mix of contextual variables that align with established risk assessment practices and can inform decision-making in the visa sponsorship process.

5.4 Model Robustness and Generalisation

The model's performance was evaluated across multiple metrics to assess robustness. The ROC-AUC score of 0.8876 indicates strong discriminative ability across different classification thresholds. The ROC curve, presented in Figure 5.4, visualises this discriminative performance by plotting the true positive rate against the false positive rate at various threshold settings. The curve demonstrates that the model maintains high true positive rates while keeping false positive rates relatively low, indicating effective separation between the two classes. The weighted F1-score of 0.9298 reflects the model's ability to maintain performance when accounting for class imbalance, while the macro F1-score of 0.6006 shows the challenge of balancing performance across both classes.

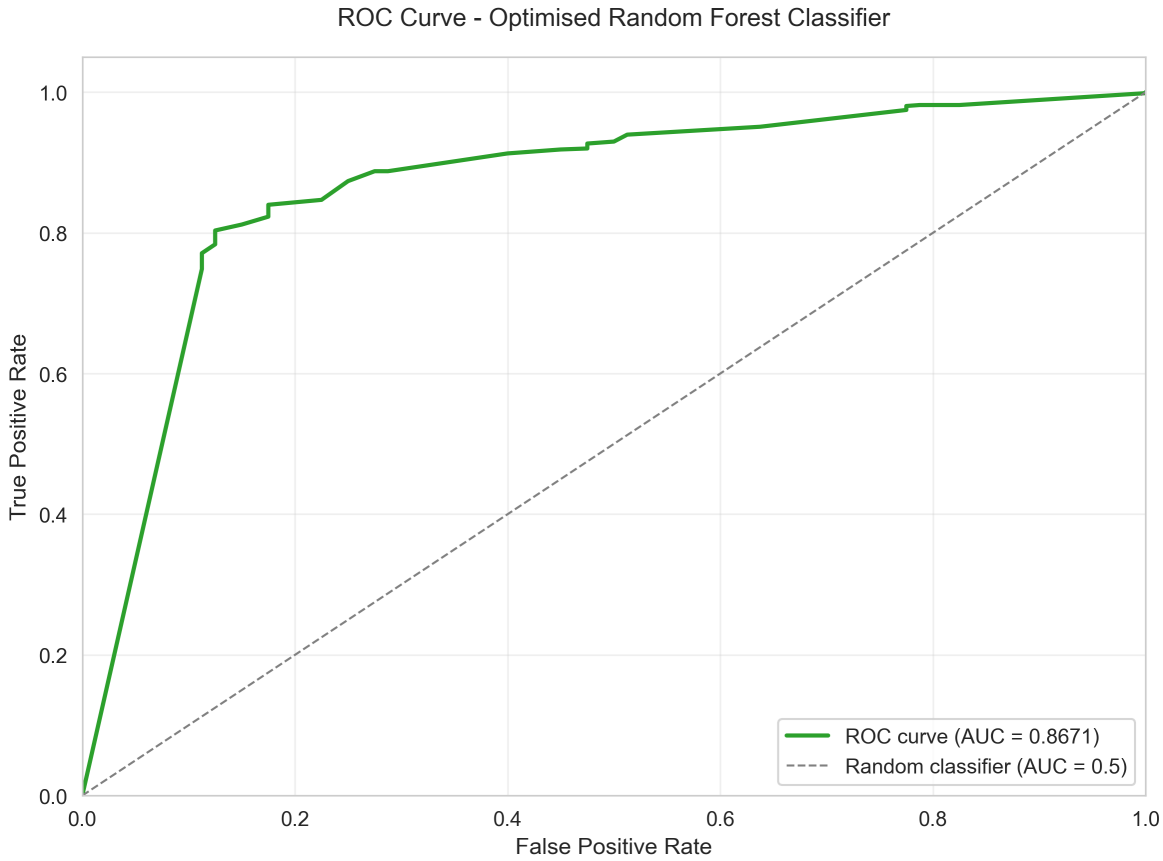


Figure 5.4: ROC curve for the optimised Random Forest classifier on the test set. The curve shows the trade-off between true positive rate (sensitivity) and false positive rate (1 - specificity) across different classification thresholds. The area under the curve (AUC) of 0.8876 indicates strong discriminative ability.

Cross-validation results on the training set demonstrated consistent performance, with a mean balanced accuracy of 0.8197 and a standard deviation of 0.0119 across five folds, indicating stable performance and low variance. This suggests that the model generalises well and is not overly sensitive to specific data partitions.

The test set performance (balanced accuracy: 0.8456) exceeded the cross-validation mean (0.8197), indicating that the model performs well on unseen data and has not overfitted to the training set. This performance improvement on the test set, while modest, suggests that the hyperparameter optimisation successfully identified a configuration that generalises effectively.

Chapter 6

Conclusions

6.1 Addressing Research Questions and Objectives

This section directly addresses the three research questions and three objectives outlined in Chapter 1, drawing upon the methodological approach, results, and findings presented throughout this dissertation.

6.1.1 Research Question 1: Data Sufficiency for Risk Classification

The first research question asked: *Is the data collected by Aston University sufficient to develop an accurate and reliable risk classification model?*

The analysis demonstrates that the available data, while not exhaustive, was sufficient to develop a functional risk classification model. The dataset comprised 3,957 records across four intake cycles (2022–2025), with 16 features relevant to credibility assessment. The model achieved a test set balanced accuracy of 0.8456 and a ROC-AUC score of 0.8876, indicating strong discriminative ability. These results suggest that the existing data collection processes at Aston University capture sufficient information to support meaningful risk classification.

However, the data sufficiency must be considered in context. The feature importance analysis revealed that a single feature accommodation status (specifically, whether accommodation information is missing) accounts for 73.17% of the model's predictive power. While this demonstrates that the available data contains strong signals, it also highlights a potential limitation: the model's performance is heavily dependent on one feature, which may not be available or reliable in all contexts. The remaining top features (age: 7.45%, Student Deposit Amount: 5.36%, Course Fee After Scholarships: 3.49%, and CourseFee: 2.29%) collectively account for 18.59% of feature importance, suggesting that while additional features contribute to classification, the model's success is primarily driven by accommodation status.

The dataset's class imbalance (8.89:1 ratio of "Passed" to "Failed") presented challenges but was successfully addressed through class weighting strategies. The missing data proportion (4.19%) was relatively small and handled through appropriate imputation

strategies, treating missingness as an informative category for categorical variables. The temporal span of four intake cycles provided sufficient diversity to support generalisation, as evidenced by the model's consistent cross-validation performance (mean balanced accuracy: 0.8197, standard deviation: 0.0119) and strong test set performance.

While the current data was sufficient to develop a proof-of-concept classifier, there are opportunities to enhance data collection to improve model performance. Additional features that could strengthen the model include: academic qualifications and their verification status, English language proficiency test scores, previous visa history, source country risk indicators, and more detailed financial documentation. The integration of such features could potentially improve predictive accuracy and reduce reliance on a single dominant feature.

6.1.2 Research Question 2: Prototype Classifier Generalisation

The second research question asked: *Can a prototype classifier be developed that generalises effectively to unseen applicant data?*

The results provide strong evidence that a generalising prototype classifier can be developed. The optimised Random Forest classifier achieved a test set balanced accuracy of 0.8456, which exceeded the cross-validation mean (0.8197) on the training set, indicating that the model generalises well to unseen data and has not overfitted. The ROC-AUC score of 0.8876 demonstrates strong discriminative ability across different classification thresholds, while the cross-validation results showed consistent performance with low variance (standard deviation: 0.0119), suggesting stable generalisation across different data partitions.

The model's performance on the test set (792 samples, representing 20% of the dataset) demonstrates its ability to handle unseen applicants effectively. The high recall for the "Failed" class (0.8889) is particularly important for a compliance risk detection system, as it ensures that the majority of actual failures are identified (Bolton and Hand, 2002). While precision for the "Failed" class is lower (0.1905), this reflects the inherent challenge of imbalanced classification and the model's prioritisation of recall over precision for risk detection purposes.

The systematic model selection and optimisation process contributed to the model's generalisation capability. The comparative evaluation of seven classification algorithms (MLP, Decision Tree, Random Forest, Gradient Boosting, Logistic Regression, K-Nearest Neighbors, and AdaBoost) identified Random Forest as the optimal base classifier, which was then refined through a two-stage hyperparameter optimisation process. The final model configuration, with 300 estimators, maximum depth of 15, and balanced subsample class weighting, represents an optimal balance between model complexity and generalisation performance.

However, it is important to note that generalisation is context-dependent. The model was trained and evaluated on data from a single institution (Aston University) across four intake cycles. While the temporal diversity and sample size support generalisation to future intakes at the same institution, generalisation to other universities or different regulatory contexts would require validation on additional datasets (Steyerberg et al., 2010). The model's performance suggests it can serve as an effective proof-of-concept

for institutional use, but broader generalisation would benefit from multi-institutional validation.

6.1.3 Research Question 3: Implementation and Benefits for Visa Compliance Strategy

The third research question asked: *Can this model be implemented into Aston University's Visa Compliance strategy and what benefit could this bring?*

The model can be implemented as a decision support tool within Aston University's visa compliance strategy, offering several significant benefits while maintaining appropriate human oversight. The classifier serves as an early-warning mechanism that can identify high-risk applicants before the credibility interview stage, enabling compliance officers to allocate resources more effectively and focus their attention on cases that require detailed review.

The primary benefit is enhanced risk detection efficiency. With a recall of 88.89% for the "Failed" class, the model can identify the vast majority of applicants who would ultimately fail their credibility interviews, allowing compliance officers to prioritise these cases for additional scrutiny. This early identification could enable proactive interventions, such as requesting additional documentation, conducting preliminary interviews, or providing targeted support to address identified risk factors before formal visa application submission (Baesens et al., 2015).

The model's interpretability, demonstrated through feature importance analysis (see Section 5.3), provides actionable insights for compliance officers. The identification of key predictive features offers clear, understandable signals that can inform decision-making, enabling officers to focus their review on the most relevant risk factors. This interpretability aligns with the ethical framework discussed in Section 6.2, ensuring that the model supports rather than replaces human judgment.

Implementation would require integration into existing compliance workflows. The model could be deployed as a risk scoring system that assigns each applicant a risk classification (high, medium, or low) based on predicted probabilities, with high-risk cases flagged for enhanced review. This integration would need to be supported by appropriate user interfaces, training for compliance officers, and monitoring systems to track model performance over time and detect potential concept drift as applicant patterns evolve (Gama et al., 2014).

The benefits extend beyond individual case assessment to strategic compliance management. By providing consistent, data-driven risk assessments, the model could help ensure uniform application of compliance standards across different officers and time periods (Kleinberg et al., 2018). This consistency is particularly valuable given the tightened regulatory environment, where the visa refusal threshold has been reduced from 10% to 5%. The model's ability to identify risk early in the process could help institutions maintain compliance with these stricter requirements while managing their international student recruitment effectively.

However, successful implementation requires careful consideration of several factors. Regular model retraining would be necessary to adapt to changing applicant patterns and

regulatory requirements (Sculley et al., 2015). Additionally, the model's performance is dependent on data quality and completeness, requiring ongoing data collection and validation processes (Schelter et al., 2018). The ethical considerations of maintaining human oversight in decision-making are discussed further in Section 6.2.

6.1.4 Achievement of Project Objectives

The three project objectives outlined in Chapter 1 have been successfully achieved:

Objective 1: Build a predictive model classifying applicants based on visa outcome or credibility interview results. This objective was achieved through the development of an optimised Random Forest classifier that predicts credibility interview outcomes ("Passed" or "Failed") based on applicant features. The model achieved a test set balanced accuracy of 0.8456 and demonstrates strong discriminative ability (ROC-AUC: 0.8876), successfully classifying applicants according to their interview outcomes.

Objective 2: Compare the performance of different ML algorithms to determine the most effective approach for risk classification. This objective was achieved through systematic comparative evaluation of seven classification algorithms: MLP, Decision Tree, Random Forest, Gradient Boosting, Logistic Regression, K-Nearest Neighbors, and AdaBoost. The evaluation used 5-fold stratified cross-validation with balanced accuracy as the primary metric, supplemented by additional performance measures. Random Forest was identified as the optimal classifier, achieving the highest test set balanced accuracy (0.7984) among the top three performers, and was subsequently selected for hyperparameter optimisation.

Objective 3: Assess model interpretability and real-world applicability in a university compliance setting. This objective was achieved through feature importance analysis (detailed in Section 5.3), which identified the most predictive features and their relative contributions to model decisions. The top features provide interpretable signals that align with established risk assessment practices (Ngai et al., 2011). The model's design as a decision support tool, rather than an automated decision-maker, ensures its applicability within a compliance context where human oversight and judgment remain essential, as discussed in Section 6.2. The high recall for the "Failed" class (88.89%) demonstrates the model's practical utility for risk detection.

6.2 Ethical Conclusions

The application of machine learning algorithms to visa application assessment raises significant ethical considerations regarding the role of automated decision-making in immigration processes. The prospect of replacing human decision-makers with algorithmic systems is controversial, as it could potentially remove the nuanced judgment, contextual understanding, and balanced decision-making that human officers bring to complex cases involving individual circumstances, cultural considerations, and exceptional situations that may not be adequately captured by predictive models (Selbst and Barocas, 2018).

This project addresses these ethical concerns by positioning the classifier as a risk identification tool rather than an automated decision-maker, as discussed in Section 6.1.3. The

model is designed to support human compliance officers by providing risk classifications and feature importance insights that can be interpreted alongside other contextual factors, rather than replacing human judgment entirely. This human-in-the-loop approach ensures that final decisions remain with trained officers who can consider the full range of factors beyond what the model captures, including individual circumstances, exceptional cases, and the broader context of each application (Selbst and Barocas, 2018).

The Visa Compliance team at Aston University regularly undertakes unconscious bias training, which complements the ethical implementation of this model. This training helps ensure that human officers are aware of potential biases in their decision-making processes, whether working independently or in conjunction with algorithmic support (Devine et al., 2012). The combination of bias-aware human judgment and interpretable machine learning outputs creates a more robust ethical framework for decision-making, where both human and algorithmic limitations are acknowledged and addressed (Binns, 2018).

By focusing on risk classification and interpretability, the model serves as a decision support system that enhances rather than replaces human expertise. This approach aligns with best practices in algorithmic decision-making (Graham, 2021), where automated systems are used to augment human capabilities while maintaining human oversight and accountability for final decisions (Amershi et al., 2019).

6.3 Limitations and Future Recommendations

While this study demonstrates the feasibility of machine learning for visa compliance risk assessment, several limitations must be acknowledged, and opportunities for future enhancement identified.

6.3.1 Dataset Limitations and Feature Context

The feature importance analysis (presented in Section 5.3) reveals that accommodation status and financial indicators (fees, deposits) dominate the model's predictions, which makes contextual sense given the nature of credibility interviews. These features are directly questioned during the interview process, where officers assess whether applicants have made genuine arrangements for accommodation and can demonstrate financial capability to support their studies. The classifier is effectively learning to identify patterns that correlate with interview outcomes, capturing the relationship between these directly-assessed factors and final decisions.

The dominance of accommodation status as the most important feature (accounting for 73.17% of feature importance) warrants particular consideration. Unlike many other features in the dataset such as date of birth, nationality, or country of application accommodation status represents one of the few variables over which applicants have direct control and which reflects their active preparation for study. This distinction is significant because it suggests that the model is capturing patterns related to applicant agency and preparedness, rather than relying primarily on immutable demographic characteristics. The accommodation feature may therefore represent an important dimension of the

interview outcomes that other features cannot capture: the extent to which applicants have demonstrated genuine commitment and preparation through concrete actions, such as securing accommodation arrangements.

This interpretation aligns with the likely unconscious implications embedded in the interview decisions being modelled. Compliance officers conducting credibility interviews may unconsciously place greater weight on evidence of active preparation and commitment manifested through accommodation arrangements than on demographic characteristics that applicants cannot control (Greenwald and Krieger, 2006). The model's strong reliance on accommodation status may thus reflect not only the explicit questions asked during interviews, but also the implicit decision-making patterns of officers who subconsciously value indicators of preparation and commitment (Kahneman, 2011). This suggests that accommodation status captures a meaningful signal about applicant credibility that goes beyond what demographic or financial features alone can provide, representing a dimension of genuine study intention that is both observable and actionable for applicants.

The context of how Pre-CAS Interviews are currently conducted at Aston University provides important insight into the data being modelled and potential limitations. Interviews are conducted by compliance officers who select students based on two primary criteria: UKVI-defined risk regions (primarily based on nationality) and course-level risk assessments, where applicants to courses considered high-risk (such as MSc Business and Management) are required to be invited regardless of nationality. This selection process means that the dataset is not representative of all applicants, but rather reflects a risk-stratified sample where certain nationalities and course choices are over-represented. The interviews themselves cover 10 sections that systematically assess applicant credibility, including questions about reasons for choosing the UK and Aston University over other options, preparedness for studies (encompassing funding, accommodation, and cultural knowledge), and justification for course selection. The model's strong reliance on features such as accommodation status and financial indicators aligns with these interview sections, suggesting that the classifier is effectively learning the patterns that officers use to assess these specific dimensions of credibility.

However, this selection process introduces potential limitations that must be acknowledged. The nationality-based selection criteria mean that the model may be learning patterns that are specific to certain risk regions, potentially limiting its generalisability to applicants from nationalities that are less frequently interviewed (Zadrozny, 2004). Similarly, the over-representation of high-risk courses in the dataset may mean that the model's predictions are more calibrated to these specific course contexts. Additionally, because interviews are only conducted for selected applicants, the model cannot learn from patterns in applicants who were not interviewed, creating a potential selection bias where the training data may not fully represent the broader applicant population (Heckman, 1979). This pattern recognition also highlights a limitation: the model's performance is constrained by the features available in the dataset, which reflect what is currently collected and assessed, rather than what might be most predictive of genuine study intentions.

The dataset available for this study is limited by the scope of features it includes. While the 16 features used provide sufficient signal for a functional classifier, the model could be significantly more robust and informative with a wider array of more indicative features.

Additional features that could enhance predictive capability include: academic qualification verification status, English language proficiency test scores and their verification, previous visa history and compliance records, source country risk indicators based on historical refusal rates, detailed financial documentation verification status, course alignment with previous academic background, and evidence of genuine study preparation (such as pre-arrival engagement with the institution). The integration of such features would not only improve model accuracy but also provide richer interpretability for compliance officers, enabling more nuanced risk assessment that considers multiple dimensions of applicant credibility.

6.3.2 Algorithmic and Model Limitations

The Random Forest classifier selected for this study provides binary classifications and feature importance scores, but could be enhanced to provide more nuanced risk indicators. While Random Forest can produce probability estimates, some alternative algorithmic approaches, such as logistic regression or calibrated neural networks, may offer more interpretable continuous risk indicators based on proximity to decision boundaries (Niculescu-Mizil and Caruana, 2005). This capability would be valuable in a compliance context, as it would enable officers to distinguish between applicants who are clearly high-risk or low-risk versus those who fall in a borderline region requiring additional scrutiny. The current model's output, while effective for proof-of-concept purposes, could be enhanced to provide more granular risk stratification.

Future work should explore the integration of algorithms that provide continuous risk scores or probability estimates, enabling the development of risk stratification systems (e.g., high, medium, low risk) rather than binary classifications. Logistic regression, with its inherent probability outputs, or calibrated neural networks could provide such capabilities. Additionally, ensemble approaches that combine multiple algorithms could leverage the strengths of different methods: tree-based models for interpretability, logistic regression for probability estimation, and neural networks for capturing complex non-linear relationships (Zhou, 2012).

6.3.3 Temporal and Scope Limitations

This study is limited by its focus on September intake cycles only, despite the university offering multiple intakes throughout the year. The exclusion of other intake periods (such as January or May intakes) means that the model may not capture seasonal patterns, temporal variations in applicant characteristics, or intake-specific risk factors that could influence credibility assessment. Different intake periods may attract different applicant profiles, have varying levels of competition for places, or be subject to different regulatory contexts at the time of application (Hand, 2006). The model's generalisation to non-September intakes, while likely reasonable given the consistency of interview processes, has not been explicitly validated.

Future research should incorporate data from all intake cycles to develop a more comprehensive model that accounts for temporal variations. This would enable the identification of intake-specific risk patterns and ensure the model's applicability across the full academic calendar. Additionally, longitudinal analysis across multiple years would help

identify concept drift—changes in applicant patterns or risk factors over time (Gama et al., 2014)—and support the development of adaptive models that can be retrained periodically to maintain performance as patterns evolve (Widmer and Kubat, 1996).

6.3.4 Future Recommendations

Based on these limitations, several recommendations emerge for future development:

Enhanced Dataset: This project used the dataset from a single compliance report which tracks the progress of international applicants in obtaining a CAS, if the dataset was enriched with other features available in the wider admissions process, it could result in a more robust and informative classifier.

Algorithmic Enhancement: Future iterations should explore algorithms that provide continuous risk scores or probability estimates, enabling more nuanced risk stratification. The development of ensemble methods that combine multiple algorithmic approaches could leverage complementary strengths and provide more robust predictions.

Multi-Intake Validation: The model should be validated on data from all intake cycles to ensure broad applicability. This would involve retraining the model on a comprehensive dataset including all intake periods and evaluating performance across different temporal contexts.

Real-Time Learning and Adaptation: Implementation of systems that can detect concept drift and support periodic model retraining would ensure the classifier remains effective as applicant patterns and regulatory contexts evolve (Gama et al., 2014). This would require establishing monitoring systems to track model performance over time and identify when retraining is necessary (Polyzotis et al., 2018).

Integration with Decision Support Systems: Future development should focus on creating user-friendly interfaces that present risk scores, feature importance, and contextual information to compliance officers in an interpretable format (Lipton, 2018). This would maximise the practical utility of the model while maintaining the human-in-the-loop approach essential for ethical implementation.

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Appendix A

Appendices

A.1 Anonymised Dataset

The anonymised dataset used in this study is available at:

`https://github.com/Alex-Carter99/Uni-Projects/blob/main/Combined.csv`

This dataset contains 3,957 anonymised records of international student applicants, with all personally identifiable information replaced with proxy identifiers as described in Chapter 3.

A.2 Final Pipeline Workbook

The complete machine learning pipeline, including data preprocessing, model selection, hyperparameter optimization, and evaluation, is available as a Jupyter notebook at:

`https://github.com/Alex-Carter99/Uni-Projects/blob/main/Final.pipeline.ipynb`

This notebook contains all code used to implement the methodology described in Chapter 4, including the preprocessing pipeline, comparative classifier testing, Random Forest hyperparameter optimization, and final model evaluation.